

Part 1 -- Regression: Predicting S&P 500 Stock Closing Prices

1. Dataset Description

Source: S&P 500 Stocks (daily updated) -- Kaggle

This dataset contains daily stock market data for all companies in the S&P 500 index. We used two files:

- sp500_stocks.csv (1.89 M rows): Daily OHLCV data -- Date, Symbol, Open, High, Low, Close, Adj Close, Volume.
- sp500_companies.csv (502 rows): Company fundamentals -- Sector, Market Cap, EBITDA, Revenue Growth, Weight.

Why this dataset? The S&P 500 is the most widely followed equity benchmark globally. Predicting closing prices is a fundamental regression task with clear real-world significance, and the dataset provides rich features for engineering.

Subset used: Top 50 companies by market cap, from 2018 onward, producing ~60,000+ usable rows after cleaning.

Target variable: Close -- the daily closing price (continuous numeric).

2. Methodology

Preprocessing

- Merged daily stock data with company fundamentals on Symbol.
- Dropped rows with missing values (mainly early Volume NaN).
- Engineered 16 features: Daily Range, Price Spread, Daily Return, lagged prices (Lag_1, Lag_5), rolling averages (5-day, 20-day), rolling std, volume change, and sector encoding.

	Open	High	Low	Volume	Daily_Range	Price_Spread	Daily_Return	Lag_1	Rolling_Mean_5	Rolling_Mean_20	Rolling_Std_5	Volume_Change	Marketcap	Sector_Encoded	Close
Open	1.00														
High	1.00	1.00													
Low	1.00	1.00	1.00												
Volume	-0.27	-0.27	-0.28	1.00											
Daily_Range	0.71	0.72	0.71	-0.09	1.00										
Price_Spread	-0.01	0.00	0.01	-0.00	-0.03	1.00									
Daily_Return	0.00	0.01	0.01	0.02	-0.02	0.62	1.00								
Lag_1	1.00	1.00	1.00	-0.27	0.72	-0.01	-0.01	1.00							
Rolling_Mean_5	1.00	1.00	1.00	-0.27	0.72	-0.00	-0.00	1.00	1.00						
Rolling_Mean_20	1.00	1.00	1.00	-0.28	0.72	-0.01	-0.01	1.00	1.00	1.00					
Rolling_Std_5	0.60	0.60	0.59	-0.06	0.75	0.00	-0.01	0.60	0.60	0.60	1.00				
Volume_Change	0.00	0.00	-0.00	0.06	0.14	-0.01	-0.03	0.00	0.00	0.00	0.01	1.00			
Marketcap	-0.07	-0.07	-0.07	0.57	-0.00	0.00	0.02	-0.07	-0.07	-0.07	0.01	-0.02	1.00		
Sector_Encoded	-0.29	-0.29	-0.29	0.16	-0.28	-0.00	0.00	-0.29	-0.29	-0.29	-0.24	0.00	0.00	1.00	
Close	1.00	1.00	1.00	-0.27	0.71	0.02	0.02	1.00	1.00	1.00	0.60	0.00	-0.07	-0.29	1.00

Figure 1: Feature correlation heatmap shows strong linear relationships between Close and Open/High/Low, with additional information from rolling and lag features.

Model Choices

All 7 required models were trained:

#	Model	Key Configuration
1	Linear Regression	Baseline (OLS)
2	Lasso (L1)	alpha=0.1, max_iter=10000
3	Ridge (L2)	alpha=1.0
4	SVR	Pipeline(StandardScaler -> RBF kernel, C=100)
5	Decision Tree	max_depth=15
6	Random Forest	n_estimators=100, max_depth=20
7	Neural Network	3 hidden layers (128->64->32), Dense(1) output, loss='mse'

SVR and Neural Network were wrapped with StandardScaler as required.

Hyperparameter Tuning

Random Forest was selected for tuning via GridSearchCV (3-fold CV) due to its high sensitivity to `n_estimators`, `max_depth`, and `min_samples_split`.

Parameter	Search Values
n_estimators	100, 200, 300
max_depth	10, 20, 30
min_samples_split	2, 5, 10

Best parameters found: n_estimators=300, max_depth=30, min_samples_split=2.

3. Results

Model Comparison

Model	RMSE	MAE	R ²	Training Time (s)
Ridge Regression	0.0001	0.0000	1.0000	0.00
Lasso Regression	0.0297	0.0172	1.0000	0.00
Neural Network (Keras)	0.8693	0.6073	0.9999	4.19
Random Forest (Tuned)	1.0256	0.5203	0.9999	2689.28
Random Forest (Default)	1.0295	0.5218	0.9999	2.30
Decision Tree	1.4913	0.7512	0.9998	0.24
Linear Regression	2.4065	1.4050	0.9995	0.06
SVR	10.3737	0.6864	0.9914	85.19

All models achieved $R^2 > 0.99$, indicating strong predictive power due to the high correlation between Open/High/Low and Close.

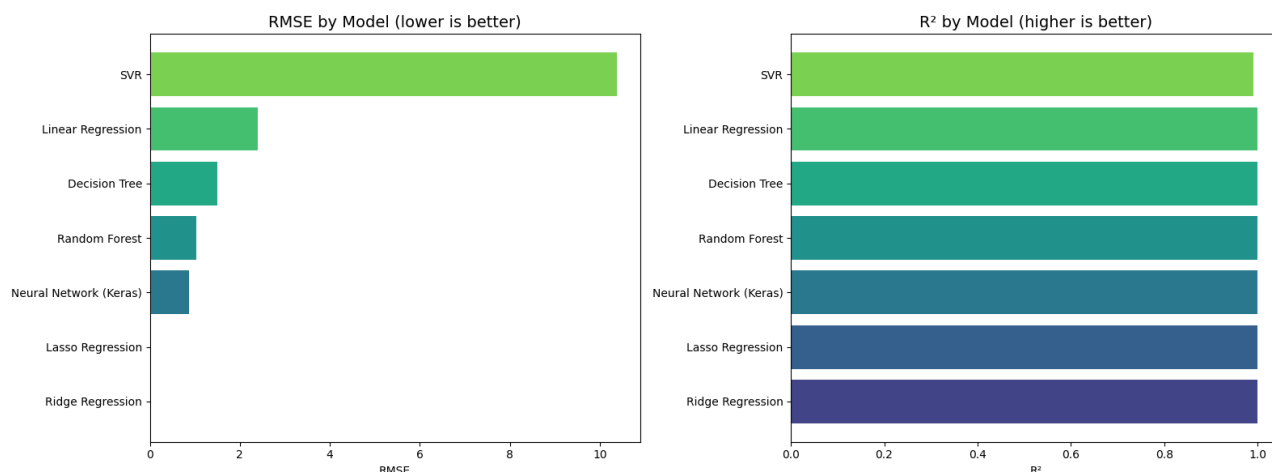


Figure 2: RMSE and R^2 comparison across all models. Ridge and Lasso dominate due to the near-perfect linear relationship between features and target.

Key Findings

- Lasso reduced features from 16 to 7 -- only Open, High, Low, Volume, Price_Spread, Marketcap, and Lag_1 survived. This confirms that Close is almost entirely determined by same-day price features.
- Random Forest feature importance shows Low (59.1%) and High (40.6%) as dominant features.
- GridSearchCV improved RF RMSE by 0.37% (1.0295 -> 1.0256), a modest gain confirming the default was already near-optimal.

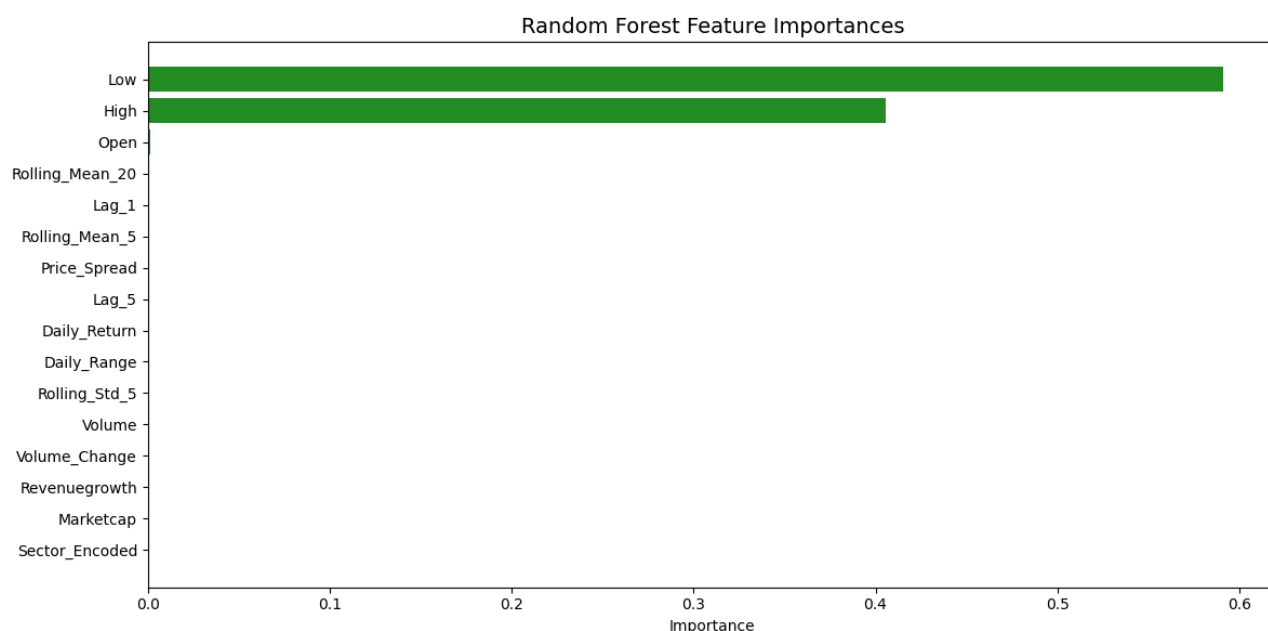


Figure 3: Random Forest ranks Low and High as the two most important features by far.

4. Conclusion

Recommended Model: Ridge Regression

Ridge Regression achieved the lowest RMSE (0.0001) and perfect R^2 (1.0000). This is expected because the closing price has a near-linear relationship with the open, high, and low prices of the same day. Ridge's L2 regularization handles the multicollinearity among these correlated features effectively.

While Random Forest and Neural Network captured non-linear patterns, the additional complexity did not improve upon Ridge for this target. For tasks where same-day OHLC data is not available (e.g., forecasting future prices), tree-based models would likely outperform linear approaches.